

Name: _____

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MyDesign Portfolio

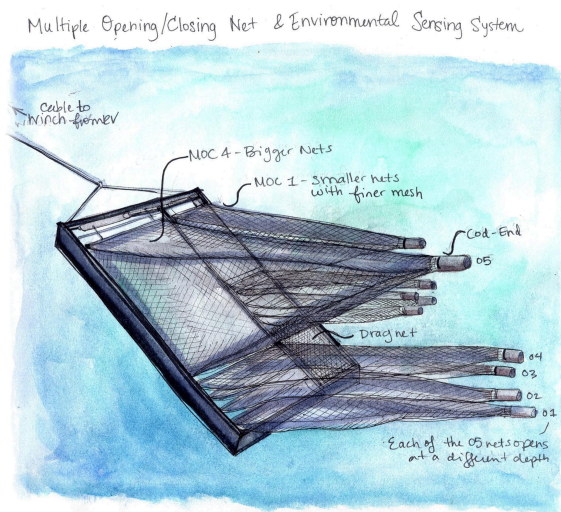
Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <MOCNESS>

<https://en.wikipedia.org/wiki/MOCNESS>

Image



MOCNESS - Plankton Up Close - Artist at Sea Part 3 — Kristin Link

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"The system is towed behind a research vessel with a speed of up to 2.5 knots (1.3 m/s) and consists of five to twenty nets with a mesh size from 64 μm to 3 mm and an area of 0.25 to 20 m² (although the last one is a 2x10 m systems) which are opened and closed computer controlled at desired depth."

Analysis

Most microplastics either sink or float. There are rarely deep water samples for microplastics this way.

Prior Design Attempt #2 - <Visual inspection with a stereomicroscope>

<https://www.sciencedirect.com/science/article/abs/pii/S007668792030375X>

Image



[How to Select the Right Solution for Visual Inspection | Science Lab | Leica Microsystems](#)

Details

"Visual sorting is often used before the spectroscopic method to save time. Large microplastic pieces (> 1 mm) can often be identified by the naked eye even though spectroscopic identification is still needed to confirm the polymer type because visual identification is very subjective. The size of the microplastic is usually measured under a stereomicroscope."

Analysis



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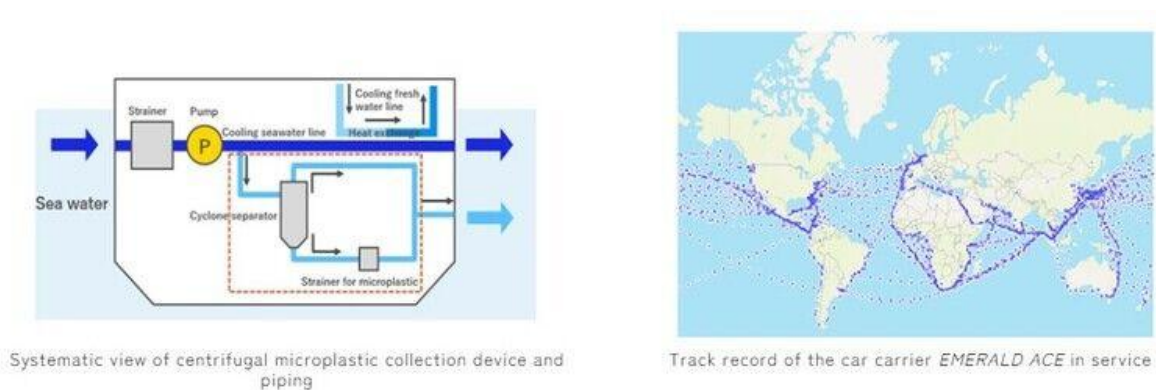
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If we use this method it would make things a lot easier, but it is not filtering out enough things so this will not be useful for our design. Our stakeholder wants a specific size for the microplastics, and they do not want to sort it themselves. This would also take more scientific equipment that we don't have, like a microscope. Even though the test is one of the better methods to use we cannot use this for our project.

Prior Design Attempt #3 - <Mitsui O.S.K Lines>

<https://www.cleantesky.com/innovation/mp-collection-device#:~:text=The%20MP%20collection%20device%20works,water%20back%20into%20the%20ocean.>

Image



Details

"The MP collection device works by straining microplastics out of concentrated seawater. It achieves this by continually drawing in seawater, which flows through a strainer and a pump that separates the plastics from the water before releasing the water back into the ocean."

Analysis

I think this device would be better for a long time usage rather than a short testing. It cannot be easily pulled out of the water or navigated around. This is a good way to clean the water, not test the microplastics.

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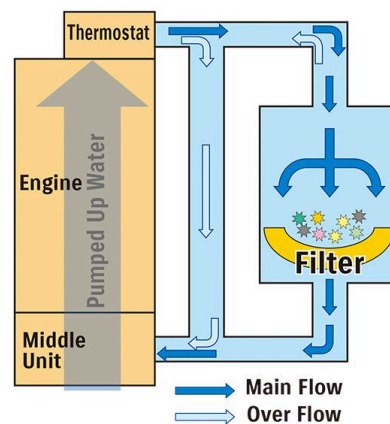
Prior Design Attempt #4 - <Suzuki Clean Ocean Project>

[https://www.globalsuzuki.com/suzuki_family/marine/suzuki announced the development of micro-plastic collecting device.html](https://www.globalsuzuki.com/suzuki_family/marine/suzuki_announced_the_development_of_micro-plastic_collecting_device.html)

Image



WITH MICRO PLASTIC COLLECTING DEVICE



Details

"To tackle these issues, we focused on the structure of the outboard motor, which pumps up tons of seawater to cool the engine and then returned to the ocean. We developed a device which collects micro-plastic waste from the returning water. Through this device, micro-plastic waste around the water surfaces can be collected just by running the boat."

Analysis

It is still being tested and fixed to this day, so it might miss some smaller microplastics since the motor is pushing it out so fast. It also does not seem very accessible since it is a whole boat motor and those are very expensive, especially ones that have features to make the water cleaner. When buying a motor, people usually think about which motor is going to make the boat go the fastest, not if they're polluting the water or not.

Prior Design Attempt #5 - Microbots

Source

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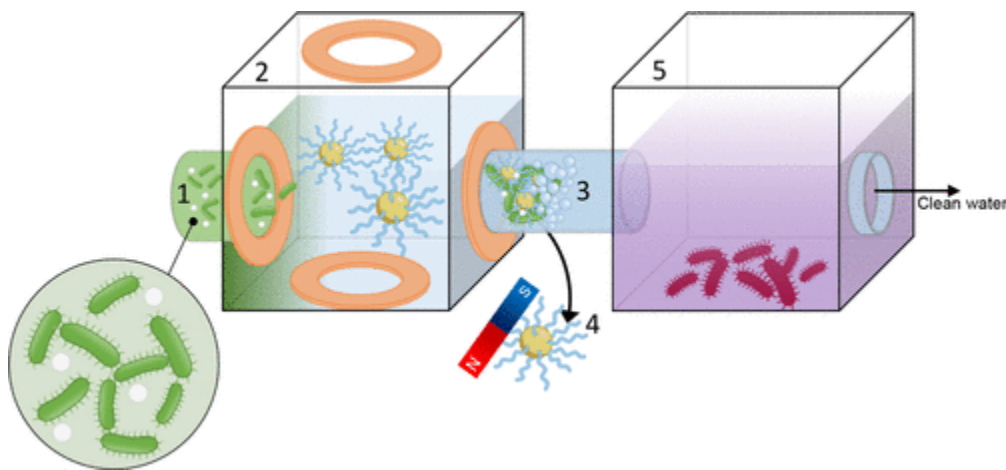
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<https://pubs.acs.org/doi/full/10.1021/acsnano.4c02115>

The microbots were made by a team at the central european institute of technology.

"In summary, this multifaceted experiment demonstrated the ability of the self-propelled rotating microrobotic planes to capture bacterial contaminants and microplastics from water. The integration of qualitative and quantitative assessments emphasizes their robust performance and potential as materials for treating various environmental pollutants simultaneously. This approach can stimulate the development of more sophisticated materials, including hybrid systems capable of capturing both positively and negatively charged contaminants at the same time. This is especially attractive for real-world water purification applications where contaminated water samples usually contain several types of pollutants." - good summarizing paragraph from the article.

Image



This is from the [article](#) cited above. There aren't many better images, mostly just diagrams because of the size.

Details

These microbots are remotely operated in a very different way. They aren't long distances and are used to collect much more specific samples from an existing water sample. Though many resources on the microbots talk about bacteria, they have a lot of application in analyzing microplastics. The microbots are controlled by changing the magnetic field that they sit in. They also have hands made out of very specific materials (the main focus of the

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paper) that allows them to grab either different bacteria or types of microplastics. The study was able to successfully take a sample of water and move specific types of microplastics into a cleaner sample, which has pretty cool implications for analysis on other samples.

Analysis

These robots solve a slightly different problem. Though they still collect microplastic samples, it's on a much smaller scale with an already isolated sample of water. It falls more or less in between collecting samples and analyzing samples. But for what it was set out to do it was extremely effective. Additionally the researcher made a method of recycling the microbots which worked really well, so they're relatively sustainable for what they do.